Report — Coursework 1

Netflix Content Strategy Analysis

Section 1: Dataset Overview and Pre-processing

1.1 Source of the Dataset

The dataset selected is the **Netflix Movies and TV Shows** dataset, publicly available on Kaggle at: https://www.kaggle.com/datasets/shivamb/netflix-shows, one of the recommended sources in the assessment brief. Compiled by Shivam Bansal from Netflix's publicly available catalogue metadata, it contains **8,807 observations and 12 attributes**, well exceeding the minimum requirements. Attributes include show_id, type, title, director, cast, country, date_added, release_year, rating, duration, listed_in, and description, covering a rich mix of categorical and numeric data suitable for machine learning analysis.

It is important to note several **dataset limitations**: the data only extends to mid-2021, so recent strategic shifts are not captured; no viewing figures are included, meaning the dataset reflects availability rather than popularity; and 1,320 titles list multiple countries in a single field, which may affect geographic analysis accuracy.

1.2 Hypothesis and Problem Statements

For this analysis, the role adopted is that of a Data Scientist working for a **media analytics consultancy** advising Netflix's content acquisition team. The central hypothesis is:

"An analysis of Netflix's content catalogue will reveal meaningful patterns in content type, audience rating, and geographic origin that can inform future acquisition and platform growth strategy."

Problem 1 — Content Type Classification: Can a classification model accurately predict whether a title is a Movie or TV Show using attributes such as rating, primary genre, release year, and year added? This supports automated content tagging and recommendation engine optimisation, with direct impact on subscriber retention.

Problem 2 — Content Rating Trends: Has the proportion of TV-MA rated content increased year-on-year between 2015 and 2021? TV-MA additions grew from 29 titles in 2015 to 738 in 2019, confirming a significant upward trend. Understanding this shift helps align commissioning decisions with audience demand and informs regional licensing strategy.

Problem 3 — Regional Production Patterns: Do distinct geographic clusters exist in Netflix's catalogue in terms of content type and genre specialisation? The US dominates with 2,815 titles, followed by India (972) and the UK (419), revealing clear gaps in African and Latin American content that represent strategic growth opportunities.

1.3 Pre-processing and Data Interpretation

Initial inspection revealed several data quality issues. Seven preprocessing steps were applied, documented fully in the accompanying Jupyter notebook:

#	Issue	Action Taken	Justification
1	Duplicate rows	Checked and removed	None found — confirms integrity

2	Misplaced duration values in rating column (e.g. '74 min')	Moved to duration; rating set to NaN	3 rows corrected — would have caused incorrect model inputs
3	Missing: director 29.91%, cast 9.37%, country 9.44%	Filled with 'Unknown'	Dropping rows would remove ~30% of data, introducing bias
4	Missing rating values (0.05%)	Imputed with mode: TV-MA	TV-MA is the most frequent rating (36.5%) — statistically appropriate
5	date_added as unformatted string	Converted to datetime; year_added and month_added extracted	String format is unusable for trend analysis or ML models
6	duration mixes incompatible units	Split into duration_minutes and duration_seasons	Without this split, duration is an unusable string in any ML model
7	listed_in contains multiple genre tags	Primary genre extracted as primary_genre	Simplifies categorical encoding while retaining the key genre signal

After preprocessing, the dataset retains **8,797 rows and 17 attributes** (12 original + 5 engineered). Key findings from exploratory analysis: Movies account for **69.6%** of titles vs **30.4%** TV Shows (class ratio 2.3:1); TV-MA is the dominant rating at **36.5%**; the US leads production with 2,815 titles; and Dramas are the most common genre (1,600 titles). This imbalance and these attribute distributions directly inform the technique selection in Section 2.

Raw Dataset — Before Pre-processing

The screenshots below show the raw dataset as downloaded from Kaggle, with visible issues including unformatted date_added values, blank director and cast cells, and mixed duration units.

	A	B	C	D	E	F	G	H	I	J	K	L
1	show_id	type	title	director	cast	country	date_added	release_year	rating	duration	listed_in	description
2	s1	Movie	Dick Johnson	Kirsten Johnson		United States	25-Sep-21	2020	PG-13	90 min	Documentari	As her father nears the end of his life, filmmaker Kirsten Johnson stages his death in inventive
3	s2	TV Show	Blood & Water		Ama Qamata	South Africa	24-Sep-21	2021	TV-MA	2 Seasons	International	After crossing paths at a party, a Cape Town teen sets out to prove whether a private-school sw
4	s3	TV Show	Ganglands	Julien Leclerc	Sami Bouajila, Tracy Gotoa		24-Sep-21	2021	TV-MA	1 Season	Crime TV Shc	To protect his family from a powerful drug lord, skilled thief Mehdi and his expert team of robl
5	s4	TV Show	Jailbirds New Orleans				24-Sep-21	2021	TV-MA	1 Season	Docuseries, F	Feuds, flirtations and toilet talk go down among the incarcerated women at the Orleans Justic
6	s5	TV Show	Kota Factory		Mayur More,	India	24-Sep-21	2021	TV-MA	2 Seasons	International	In a city of coaching centers known to train Indiaâ€™s finest collegiate minds, an earnest but u
7	s6	TV Show	Midnight Ma	Mike Flanaga	Kate Siegel, Zach Gilford, H		24-Sep-21	2021	TV-MA	1 Season	TV Dramas, T	The arrival of a charismatic young priest brings glorious miracles, ominous mysteries and rene
8	s7	Movie	My Little Pon	Robert Culler	Vanessa Hudgens, Kimiko		24-Sep-21	2021	PG	91 min	Children & Fa	Equestria's divided. But a bright-eyed hero believes Earth Ponies, Pegasi and Unicorns should b
9	s8	Movie	Sankofa	Haile Gerima	Kofi Ghanaba	United States	24-Sep-21	1993	TV-MA	125 min	Dramas, Inde	On a photo shoot in Ghana, an American model slips back in time, becomes enslaved on a plar
10	s9	TV Show	The Great Bri	Andy Devons	Mel Giedroyc	United Kingd	24-Sep-21	2021	TV-14	9 Seasons	British TV Sh	A talented batch of amateur bakers face off in a 10-week competition, whipping up their best
11	s10	Movie	The Starling	Theodore Me	Melissa McC	United States	24-Sep-21	2021	PG-13	104 min	Comedies, Dr	A woman adjusting to life after a loss contends with a feisty bird that's taken over her garden â
12	s11	TV Show	Vendetta: Truth, Lies and The Mafia				24-Sep-21	2021	TV-MA	1 Season	Crime TV Shc	Sicily boasts a bold "Anti-Mafia" coalition. But what happens when those trying to bring down
13	s12	TV Show	Bangkok Bre	Kongkiat Kon	Sukollawat Kanarot, Susha		23-Sep-21	2021	TV-MA	1 Season	Crime TV Shc	Struggling to earn a living in Bangkok, a man joins an emergency rescue service and realizes he
14	s13	Movie	Je Suis Karl	Christian Sch	Luna Wedler, Germany, Cz		23-Sep-21	2021	TV-MA	127 min	Dramas, Inte	After most of her family is murdered in a terrorist bombing, a young woman is unknowingly lu
15	s14	Movie	Confessions	Bruno Garott	Klara Castanho, Lucca Pico		22-Sep-21	2021	TV-PG	91 min	Children & Fa	When the clever but socially-awkward TetÃª joins a new school, she'll do anything to fit in. But
16	s15	TV Show	Crime Stories: India Detectives				22-Sep-21	2021	TV-MA	1 Season	British TV Sh	Cameras following Bengaluru police on the job offer a rare glimpse into the complex and chall
17	s16	TV Show	Dear White People		Logan Brown	United States	22-Sep-21	2021	TV-MA	4 Seasons	TV Comedies	Students of color navigate the daily slights and slippery politics of life at an Ivy League college
18	s17	Movie	Europe's Mo	Pedro de Echave	GarcÃa, Pablo AzorÃn		22-Sep-21	2020	TV-MA	67 min	Documentari	Declassified documents reveal the post-WWII life of Otto Skorzeny, a close Hitler ally who esca
19	s18	TV Show	Falsa identidad		Luis Ernesto	Mexico	22-Sep-21	2020	TV-MA	2 Seasons	Crime TV Shc	Strangers Diego and Isabel flee their home in Mexico and pretend to be a married couple to es
20	s19	Movie	Intrusion	Adam Salky	Freida Pinto, Logan Marsh		22-Sep-21	2021	TV-14	94 min	Thrillers	After a deadly home invasion at a coupleâ€™s new dream house, the traumatized wife search
21	s20	TV Show	Jaguar		Blanca SuÃ¡rez, IvÃ¡n Mar		22-Sep-21	2021	TV-MA	1 Season	International	In the 1960s, a Holocaust survivor joins a group of self-trained spies who seek justice against N
22	s21	TV Show	Monsters Ins	Olivier Megaton			22-Sep-21	2021	TV-14	1 Season	Crime TV Shc	In the late 1970s, an accused serial rapist claims multiple personalities control his behavior, se
23	s22	TV Show	Resurrection: Ertugrul		Engin Altan E	Turkey	22-Sep-21	2018	TV-14	5 Seasons	International	When a good deed unwittingly endangers his clan, a 13th-century Turkish warrior agrees to fig
24	s23	Movie	Avvai Shanm	K.S. Ravikum	Kamal Hassan, Meena, Ge		21-Sep-21	1996	TV-PG	161 min	Comedies, In	Newly divorced and denied visitation rights with his daughter, a doting father disguises himsel

Figure 1: Raw dataset rows 1–29. Unformatted date_added, missing cast values, and mixed duration formats.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R
214	s213	TV Show	Rebellion		Charlie Murp	Ireland	27-Aug-21	2016	TV-MA	2 Seasons	International	As World War I rages, three women and their families in Dublin choose sides in the violent Ea						
215	s214	TV Show	RIDE ON TIME		King & Prince, Hey! Say! JU		27-Aug-21	2021	TV-PG	3 Seasons	Docuseries, I	Take a deep dive into the beautiful world of Japan's top male idol groups from number one pr						
216	s215	Movie	SAS: Rise of t	Magnus Mar	Sam Heughan, Ruby Rose,		27-Aug-21	2021	R	124 min	Action & Adv	A special forces operative traveling from London to Paris with his girlfriend takes action when						
217	s216	Movie	Shootout at I	Apoorva Lakl	Amitabh Bac	India	27-Aug-21	2007	TV-MA	116 min	Action & Adv	Based on a true story, this action film follows an incident that stunned a nation in the early 19						
218	s217	Movie	Shor In the C	Raj Nidimoru	Sendhil Ram	India	27-Aug-21	2011	TV-14	106 min	Comedies, Di	When three small-time Mumbai crooks steal a bag on a train, they find that it's filled with we						
219	s218	Movie	The Dirty Pic	Milan Luthria	Vidya Balan,	India	27-Aug-21	2011	TV-14	145 min	Comedies, Di	After running away from home in search of movie stardom, a village girl rises to become a pro						
220	s219	TV Show	Titletown High				27-Aug-21	2021	TV-14	1 Season	Reality TV, Te	In a Georgia town where football rules and winning is paramount, a high school team tackles						
221	s220	TV Show	EDENS ZERO		Takuma Tera	Japan	26-Aug-21	2021	TV-14	1 Season	Anime Series	Aboard the Edens Zero, a lonely boy with the ability to control gravity embarks on an adventu						
222	s221	TV Show	Family Reunion		Loretta Devir	United States	26-Aug-21	2021	TV-PG	4 Seasons	Kids' TV, TV C	When the McKellan family moves from Seattle to small-town Georgia, life down South â€" an						
223	s222	Movie	Bob Ross: Ha	Joshua RoÃ£	Bob Ross		25-Aug-21	2021	TV-14	93 min	Documentari	Bob Ross brought joy to millions as the worldâ€™s most famous art instructor. But a battle for						
224	s223	TV Show	Clickbait	Brad Anderso	Zoe Kazan, Betty Gabriel, A		25-Aug-21	2021	TV-MA	1 Season	Crime TV Sho	When family man Nick Brewer is abducted in a crime with a sinister online twist, those closes						
225	s224	TV Show	John of God:	Mauricio Dias,	Tatiana Villela		25-Aug-21	2021	TV-MA	1 Season	Crime TV Sho	Idolized medium JoÃ£o Teixeira de Faria rises to international fame before horrifying abuse is						
226	s225	TV Show	Motel Makeover				25-Aug-21	2021	TV-14	1 Season	International	Amid project pitfalls and a pandemic, besties-turned-business partners bring their design mag						
227	s226	TV Show	Open Your Eyes		Maria Wawreniuk, Ignacy I		25-Aug-21	2021	TV-MA	1 Season	International	After a tragic accident, an amnesiac teen tries to rebuild her life at a memory disorders center						
228	s227	TV Show	Post Mortem: No One Dies		Kathrine Thorborg Johanse		25-Aug-21	2021	TV-MA	1 Season	International	She's back from the dead and has a newfound thirst for blood. Meanwhile, her family's funera						
229	s228	Movie	Really Love	Angel Kristi V	Kofi Siriboe,	United States	25-Aug-21	2020	TV-MA	95 min	Dramas, Inde	A rising Black painter tries to break into a competitive art world while balancing an unexpecte						
230	s229	Movie	The Novemb	Roger Donak	Pierce Brosn	United States	25-Aug-21	2014	R	108 min	Action & Adv	An ex-CIA agent emerges from retirement to protect an important witness, but he soon disco						
231	s230	Movie	The Old Way	Christopher /	Brigitte Kali C	United States	25-Aug-21	2020	TV-MA	90 min	Horror Movie	A reporter visits her birthplace in Veracruz for a story about tribal culture, only to be kidnapp						
232	s231	Movie	The River Ru	Rush Sturges			25-Aug-21	2021	TV-MA	86 min	Documentari	In this documentary, a kayaker sets out to become the first man to paddle the four great river						
233	s232	Movie	The Water M	David Oyelov	David Oyelov	United States	25-Aug-21	2021	PG	92 min	Children & Fa	Desperate to save his ailing mother, 11-year-old Gunner runs away from home on a quest to f						
234	s233	TV Show	Wheel of Fortune		Pat Sajak, Va	United States	25-Aug-21	2019	TV-G	3 Seasons	Reality TV	Pat Sajak and Vanna White host one of TV's most popular, long-running game shows, where p						
235	s234	Movie	Count Me In	Mark Lo		United Kingd	24-Aug-21	2021	TV-MA	82 min	Documentari	This documentary features some of rock's greatest drummers as they come together in an ins						
236	s235	TV Show	Oggy Oggy				24-Aug-21	2021	TV-Y	1 Season	Kids' TV, TV C	Join adorable kitten Oggy and his cast of cat pals in a bright and colorful kitty world. They're a						
237	s236	Movie	Untold: Caith	Crystal Moselle			24-Aug-21	2021	TV-PG	70 min	Documentari	Caitlyn Jenner's unlikely path to Olympic glory was inspirational. But her more challenging roa						

Figure 2: Raw dataset rows 213–242. Missing director entries and inconsistent duration units.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R
312	s311	TV Show	Cocaine Cow	Billy Corben		United States	4-Aug-21	2021	TV-MA	1 Season	Crime TV Sho	Two childhood friends go from high school dropouts to the most powerful drug kingpins in Mi						
313	s312	TV Show	Control Z		Ana Valeria B	Mexico	4-Aug-21	2021	TV-MA	2 Seasons	International	When a hacker begins releasing students' secrets to the entire high school, the socially isolate						
314	s313	TV Show	Cooking With Paris		Paris Hilton	United States	4-Aug-21	2021	TV-MA	1 Season	Reality TV	With an anyone-can-cook attitude, Paris Hilton adds her own flair to every dish. Fun is just one						
315	s314	TV Show	Autumn's Concerto		Ady An, Vann	Taiwan	3-Aug-21	2009	TV-14	1 Season	International	After losing all memories of his past, a wealthy lawyer travels to a rural village where a woman						
316	s315	Movie	Heavy	James Mange	Pruitt Taylor	United States	3-Aug-21	1995	R	104 min	Dramas, Inde	Overweight Victor hides from the world making pizzas in his mother's gloomy dive until a new						
317	s316	Movie	Ije: The Journ	Chineze Anya	Genevieve N	United States	3-Aug-21	2010	TV-MA	106 min	Dramas, Inte	When her sister is accused of multiple murders, a Nigerian woman travels to Los Angeles to ur						
318	s317	TV Show	Office Girls	Hsu Fu-chun	Roy Chiu, Ali	Taiwan	3-Aug-21	2011	TV-14	1 Season	International	A department store mogul has his son work incognito in a menial job to prove his worthiness,						
319	s318	Movie	Pray Away	Kristine Stolakis		United States	3-Aug-21	2021	PG-13	102 min	Documentari	Ex-leaders and a survivor of the so-called "conversion therapy" movement speak out about its						
320	s319	Movie	Shiny_Flakes	Eva MÃ¼ller,	Maximilian Schmidt		3-Aug-21	2021	TV-MA	97 min	Documentari	Max S. reveals how he built a drug empire from his childhood bedroom in this story that inspir						
321	s320	TV Show	The Prince Who Turns into		Joe Chen, Mi	Taiwan	3-Aug-21	2005	TV-14	1 Season	International	Before his wedding, a cold CEO is hit by a car and awakens a changed man. As he and the drive						
322	s321	TV Show	Top Secret UFO Projects: Declassified				3-Aug-21	2021	TV-PG	1 Season	Docuseries	Though claims of extraterrestrial encounters have long been dismissed, many believe the exist						
323	s322	TV Show	Two Fathers		Weber Yang,	Taiwan	3-Aug-21	2013	TV-14	1 Season	International	When two single guys learn a woman they both slept with disappeared after giving birth, they						
324	s323	TV Show	You're My Destiny		Joe Chen, Ett	Taiwan	3-Aug-21	2008	TV-MA	1 Season	International	A young woman's romantic cruise ends in a twist when she gets pregnant after a one-night sta						
325	s324	TV Show	30 Rock		Tina Fey, Ale	United States	1-Aug-21	2012	TV-14	7 Seasons	TV Comedies	Liz Lemon juggles the demands of running an NBC sketch comedy show with trying to fix her d						
326	s325	TV Show	44 Cats		Sarah Natoch	Italy	1-Aug-21	2020	TV-Y7	3 Seasons	Kids' TV	Paw-esome tales abound when singing furry friends Lampo, Milady, Pilou and Meatball band t						
327	s326	Movie	Beethoven	Brian Levant	Charles Grod	United States	1-Aug-21	1992	PG	87 min	Children & Fa	A father reluctantly agrees to let his children keep a St. Bernard puppy, but the playful pooch s						
328	s327	Movie	Beethoven's	Rod Daniel	Charles Grod	United States	1-Aug-21	1993	PG	88 min	Children & Fa	After St. Bernard Beethoven sires a brood with Missy, he and the Newton kids must intervene						
329	s328	Movie	Beowulf	Robert Zeme	Ray Winstone	United States	1-Aug-21	2007	PG-13	114 min	Action & Adv	This deftly animated take on a legendary Old English epic tells the timeless tale of a Scandinavi						
330	s329	Movie	Boyka: Undis	Todor Chapka	Scott Adkins, Bulgaria, Uni		1-Aug-21	2016	R	90 min	Action & Adv	The fourth film in the action franchise finds Boyka taking on brutal gangsters to protect the wi						
331	s330	Movie	Catch Me If Y	Steven Spielb	Leonardo DiC	United States	1-Aug-21	2002	PG-13	142 min	Dramas	An FBI agent makes it his mission to put cunning con man Frank Abagnale Jr. behind bars. But						
332	s331	Movie	Cloudy with a	Phil Lord, Chr	Bill Hader, Ar	United States	1-Aug-21	2009	PG	90 min	Children & Fa	When inventor Flint Lockwood makes clouds rain food, the citizens of Chewandswallow can fe						
333	s332	TV Show	Darwinâ€™s Game		Yusuke Kobayashi, Reina U		1-Aug-21	2020	TV-MA	1 Season	Anime Series	High schooler Kaname activates a mysterious mobile app and unwittingly joins a game pitting						
334	s333	Movie	Deep Blue Se	Renny Harlin	Thomas Jane	United States	1-Aug-21	1999	R	105 min	Action & Adv	Scientists conduct research on sharks in search of an Alzheimer's cure. But a dangerous shortc						
335	s334	Movie	Ferris Bueller	John Hughes	Matthew Bro	United States	1-Aug-21	1986	PG-13	103 min	Classic Movie	After faking an illness to skip school, popular teen Ferris Bueller enjoys a fun-filled day in Chica						

Figure 3: Raw dataset rows 311–340. Continued data quality issues across the dataset.

Cleaned Dataset — After Pre-processing

The screenshots below show the cleaned dataset with correctly formatted dates, missing values replaced, duration split into numeric columns, and five new engineered features visible.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q
1	show_id	type	title	director	cast	country	late_adde	elease_yea	rating	duration	listed_in	descriptior	year_adde	onth_adde	ation_min	ation_seas	primary_genre
2	s1	Movie	Dick Johns	Kirsten Joh	Unknown	United Sta	########	2020	PG-13	90 min	Document	As her fath	2021	9	90		Documentaries
3	s2	TV Show	Blood & W	Unknown	Ama Qama	South Afric	########	2021	TV-MA	2 Seasons	Internatior	After cross	2021	9		2	International TV Shows
4	s3	TV Show	Ganglands	Julien Lecl	Sami Boua	Unknown	########	2021	TV-MA	1 Season	Crime TV S	To protect	2021	9		1	Crime TV Shows
5	s4	TV Show	Jailbirds N	Unknown	Unknown	Unknown	########	2021	TV-MA	1 Season	Docuseries	Feuds, flirt	2021	9		1	Docuseries
6	s5	TV Show	Kota Facto	Unknown	Mayur Mo	India	########	2021	TV-MA	2 Seasons	Internatior	In a city of	2021	9		2	International TV Shows
7	s6	TV Show	Midnight M	Mike Flana	Kate Siege	Unknown	########	2021	TV-MA	1 Season	TV Dramas	The arrival	2021	9		1	TV Dramas
8	s7	Movie	My Little P	Robert Cul	Vanessa H	Unknown	########	2021	PG	91 min	Children &	Equestria's	2021	9	91		Children & Family Movies
9	s8	Movie	Sankofa	Haile Gerir	Kofi Ghana	United Sta	########	1993	TV-MA	125 min	Dramas, In	On a photo	2021	9	125		Dramas
10	s9	TV Show	The Great	Andy Devo	Mel Giedro	United Kin	########	2021	TV-14	9 Seasons	British TV S	A talented	2021	9		9	British TV Shows
11	s10	Movie	The Starlin	Theodore	Melissa M	United Sta	########	2021	PG-13	104 min	Comedies,	A woman	2021	9	104		Comedies
12	s11	TV Show	Vendetta:	Unknown	Unknown	Unknown	########	2021	TV-MA	1 Season	Crime TV S	Sicily boas	2021	9		1	Crime TV Shows
13	s12	TV Show	Bangkok B	Kongkiat K	Sukollawa	Unknown	########	2021	TV-MA	1 Season	Crime TV S	Struggling	2021	9		1	Crime TV Shows
14	s13	Movie	Je Suis Kar	Christian S	Luna Wedl	Germany,	########	2021	TV-MA	127 min	Dramas, In	After most	2021	9	127		Dramas
15	s14	Movie	Confessior	Bruno Gar	Klara Cast	Unknown	########	2021	TV-PG	91 min	Children &	When the	2021	9	91		Children & Family Movies
16	s15	TV Show	Crime Stor	Unknown	Unknown	Unknown	########	2021	TV-MA	1 Season	British TV S	Cameras f	2021	9		1	British TV Shows
17	s16	TV Show	Dear Whit	Unknown	Logan Bro	United Sta	########	2021	TV-MA	4 Seasons	TV Comed	Students o	2021	9		4	TV Comedies
18	s17	Movie	Europe's N	Pedro de E	Unknown	Unknown	########	2020	TV-MA	67 min	Document	Declassifie	2021	9	67		Documentaries
19	s18	TV Show	Falsa ident	Unknown	Luis Ernest	Mexico	########	2020	TV-MA	2 Seasons	Crime TV S	Strangers	2021	9		2	Crime TV Shows
20	s19	Movie	Intrusion	Adam Salk	Freida Pint	Unknown	########	2021	TV-14	94 min	Thrillers	After a dea	2021	9	94		Thrillers
21	s20	TV Show	Jaguar	Unknown	Blanca Suá	Unknown	########	2021	TV-MA	1 Season	Internatior	In the 1960	2021	9		1	International TV Shows
22	s21	TV Show	Monsters I	Olivier Me	Unknown	Unknown	########	2021	TV-14	1 Season	Crime TV S	In the late	2021	9		1	Crime TV Shows
23	s22	TV Show	Resurrecti	Unknown	Engin Altar	Turkey	########	2018	TV-14	5 Seasons	Internatior	When a go	2021	9		5	International TV Shows
24	s23	Movie	Avvai Shan	K.S. Raviku	Kamal Has	Unknown	########	1996	TV-PG	161 min	Comedies,	Newly divo	2021	9	161		Comedies
25	s24	Movie	Go! Go! Co	Alex Woo,	Maisie Ber	Unknown	########	2021	TV-Y	61 min	Children &	From arca	2021	9	61		Children & Family Movies
26	s25	Movie	Jeans	S. Shankar	Prashanth	India	########	1998	TV-14	166 min	Comedies,	When the	2021	9	166		Comedies
27	s26	TV Show	Love on th	Unknown	Brooke Sa	Australia	########	2021	TV-14	2 Seasons	Docuseries	Finding lov	2021	9		2	Docuseries
28	s27	Movie	Minsara Ka	Rajiv Men	Arvind Swa	Unknown	########	1997	TV-PG	147 min	Comedies,	A tangled l	2021	9	147		Comedies
29	s28	Movie	Grown Up	Dennis Du	Adam San	United Sta	########	2010	PG-13	103 min	Comedies	Mourning	2021	9	103		Comedies
30	s29	Movie	Dark Skies	Scott Stew	Keri Russel	United Sta	########	2013	PG-13	97 min	Horror Mo	A family's	2021	9	97		Horror Movies

Figure 4: Cleaned dataset rows 1–29. Formatted dates, 'Unknown' replacing missing entries, engineered columns visible.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q
214	s213	TV Show	Rebellion	Unknown	Charlie Mu	Ireland	########	2016	TV-MA	2 Seasons	Internatio	As World War I rages, three v	2021	8		2	International TV Shows
215	s214	TV Show	RIDE ON T	Unknown	King & Prin	Unknown	########	2021	TV-PG	3 Seasons	Docuserie	Take a deep dive into the bea	2021	8		3	Docuseries
216	s215	Movie	SAS: Rise c	Magnus M	Sam Heug	Unknown	########	2021	R	124 min	Action & A	A special forces operative tra	2021	8	124		Action & Adventure
217	s216	Movie	Shootout a	Apoorva L	Amitabh B	India	########	2007	TV-MA	116 min	Action & A	Based on a true story, this ac	2021	8	116		Action & Adventure
218	s217	Movie	Shor In the	Raj Nidimc	Sendhil Ra	India	########	2011	TV-14	106 min	Comedies,	When three small-time Mum	2021	8	106		Comedies
219	s218	Movie	The Dirty F	Milan Luth	Vidya Bala	India	########	2011	TV-14	145 min	Comedies,	After running away from hon	2021	8	145		Comedies
220	s219	TV Show	Titletown	Unknown	Unknown	Unknown	########	2021	TV-14	1 Season	Reality TV,	In a Georgia town where foo	2021	8		1	Reality TV
221	s220	TV Show	EDENS ZEF	Unknown	Takuma Te	Japan	########	2021	TV-14	1 Season	Anime Seri	Aboard the Edens Zero, a lon	2021	8		1	Anime Series
222	s221	TV Show	Family Reu	Unknown	Loretta De	United Sta	########	2021	TV-PG	4 Seasons	Kids' TV, T	When the McKellan family m	2021	8		4	Kids' TV
223	s222	Movie	Bob Ross:	Joshua Ro	Bob Ross	Unknown	########	2021	TV-14	93 min	Document	Bob Ross brought joy to milli	2021	8	93		Documentaries
224	s223	TV Show	Clickbait	Brad Ande	Zoe Kazan	Unknown	########	2021	TV-MA	1 Season	Crime TV S	When family man Nick Brew	2021	8		1	Crime TV Shows
225	s224	TV Show	John of Go	Mauricio D	Unknown	Unknown	########	2021	TV-MA	1 Season	Crime TV S	Idolized medium João Teixei	2021	8		1	Crime TV Shows
226	s225	TV Show	Motel Mal	Unknown	Unknown	Unknown	########	2021	TV-14	1 Season	Internatio	Amid project pitfalls and a pa	2021	8		1	International TV Shows
227	s226	TV Show	Open Your	Unknown	Maria Wa	Unknown	########	2021	TV-MA	1 Season	Internatio	After a tragic accident, an an	2021	8		1	International TV Shows
228	s227	TV Show	Post Mort	Unknown	Kathrine T	Unknown	########	2021	TV-MA	1 Season	Internatio	She's back from the dead and	2021	8		1	International TV Shows
229	s228	Movie	Really Lov	Angel Krist	Kofi Siribo	United Sta	########	2020	TV-MA	95 min	Dramas, In	A rising Black painter tries to	2021	8	95		Dramas
230	s229	Movie	The Nover	Roger Don	Pierce Bro	United Sta	########	2014	R	108 min	Action & A	An ex-CIA agent emerges fro	2021	8	108		Action & Adventure
231	s230	Movie	The Old W	Christophe	Brigitte Ka	United Sta	########	2020	TV-MA	90 min	Horror Mo	A reporter visits her birthplac	2021	8	90		Horror Movies
232	s231	Movie	The River I	Rush Sturg	Unknown	Unknown	########	2021	TV-MA	86 min	Document	In this documentary, a kayak	2021	8	86		Documentaries
233	s232	Movie	The Water	David Oye	David Oye	United Sta	########	2021	PG	92 min	Children &	Desperate to save his ailing r	2021	8	92		Children & Family Movies
234	s233	TV Show	Wheel of F	Unknown	Pat Sajak,	United Sta	########	2019	TV-G	3 Seasons	Reality TV	Pat Sajak and Vanna White h	2021	8		3	Reality TV
235	s234	Movie	Count Me	Mark Lo	Unknown	United Kin	########	2021	TV-MA	82 min	Document	This documentary features s	2021	8	82		Documentaries
236	s235	TV Show	Oggy Oggy	Unknown	Unknown	Unknown	########	2021	TV-Y	1 Season	Kids' TV, T	Join adorable kitten Oggy and	2021	8		1	Kids' TV
237	s236	Movie	Untold: Ca	Crystal Mc	Unknown	Unknown	########	2021	TV-PG	70 min	Document	Caitlyn Jenner's unlikely path	2021	8	70		Documentaries
238	s237	Movie	Boomika	Rathindrar	Aishwarya	Unknown	########	2021	TV-14	122 min	Horror Mo	Paranormal activity at a lush	2021	8	122		Horror Movies
239	s238	Movie	Boomika (	Rathindrar	Aishwarya	Unknown	########	2021	TV-14	122 min	Horror Mo	Paranormal activity at a lush	2021	8	122		Horror Movies
240	s239	Movie	Boomika (	Rathindrar	Aishwarya	Unknown	########	2021	TV-14	122 min	Horror Mo	Paranormal activity at a lush	2021	8	122		Horror Movies
241	s240	Movie	Boomika (	Rathindrar	Aishwarya	Unknown	########	2021	TV-14	122 min	Horror Mo	Paranormal activity at a lush	2021	8	122		Horror Movies
242	s241	Movie	The Witch	Han Kwang	Theo Jame	Unknown	########	2021	TV-MA	84 min	Action & A	Escaping from poverty to bec	2021	8	84		Action & Adventure
243	s242	TV Show	Manifest	Unknown	Melissa Ro	United Sta	########	2021	TV-14	3 Seasons	TV Dramas	When a plane mysteriously la	2021	8		3	TV Dramas

Figure 5: Cleaned dataset rows 213–242. Zero missing values, consistent structure throughout.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q
312	s311	TV Show	Cocaine Co	Billy Corbe	Unknown	United Sta	########	2021	TV-MA	1 Season	Crime TV S	Two childhood friends go fro	2021	8		1	Crime TV Shows
313	s312	TV Show	Control Z	Unknown	Ana Valeri	Mexico	########	2021	TV-MA	2 Seasons	Internatio	When a hacker begins releas	2021	8		2	International TV Shows
314	s313	TV Show	Cooking W	Unknown	Paris Hilto	United Sta	########	2021	TV-MA	1 Season	Reality TV	With an anyone-can-cook at	2021	8		1	Reality TV
315	s314	TV Show	Autumn's (	Unknown	Ady An, Va	Taiwan	########	2009	TV-14	1 Season	Internatio	After losing all memories of l	2021	8		1	International TV Shows
316	s315	Movie	Heavy	James Ma	Pruitt Tayl	United Sta	########	1995	R	104 min	Dramas, In	Overweight Victor hides fron	2021	8	104		Dramas
317	s316	Movie	lje: The Jo	Chineze Ar	Genevieve	United Sta	########	2010	TV-MA	106 min	Dramas, In	When her sister is accused o	2021	8	106		Dramas
318	s317	TV Show	Office Girl:	Hsu Fu-ch	Roy Chiu, A	Taiwan	########	2011	TV-14	1 Season	Internatio	A department store mogul ha	2021	8		1	International TV Shows
319	s318	Movie	Pray Away	Kristine St	Unknown	United Sta	########	2021	PG-13	102 min	Document	Ex-leaders and a survivor of	2021	8	102		Documentaries
320	s319	Movie	Shiny_Flak	Eva Müller	Maximiliar	Unknown	########	2021	TV-MA	97 min	Document	Max S. reveals how he built a	2021	8	97		Documentaries
321	s320	TV Show	The Prince	Unknown	Joe Chen,	Taiwan	########	2005	TV-14	1 Season	Internatio	Before his wedding, a cold CI	2021	8		1	International TV Shows
322	s321	TV Show	Top Secret	Unknown	Unknown	Unknown	########	2021	TV-PG	1 Season	Docuserie	Though claims of extraterres	2021	8		1	Docuseries
323	s322	TV Show	Two Fathe	Unknown	Weber Yar	Taiwan	########	2013	TV-14	1 Season	Internatio	When two single guys learn a	2021	8		1	International TV Shows
324	s323	TV Show	You're My	Unknown	Joe Chen,	Taiwan	########	2008	TV-MA	1 Season	Internatio	A young woman's romantic c	2021	8		1	International TV Shows
325	s324	TV Show	30 Rock	Unknown	Tina Fey, A	United Sta	########	2012	TV-14	7 Seasons	TV Comed	Liz Lemon juggles the deman	2021	8		7	TV Comedies
326	s325	TV Show	44 Cats	Unknown	Sarah Nat	Italy	########	2020	TV-Y7	3 Seasons	Kids' TV	Paw-esome tales abound wh	2021	8		3	Kids' TV
327	s326	Movie	Beethover	Brian Leva	Charles Gr	United Sta	########	1992	PG	87 min	Children &	A father reluctantly agrees to	2021	8	87		Children & Family Movies
328	s327	Movie	Beethover	Rod Danie	Charles Gr	United Sta	########	1993	PG	88 min	Children &	After St. Bernard Beethoven	2021	8	88		Children & Family Movies
329	s328	Movie	Beowulf	Robert Zer	Ray Winst	United Sta	########	2007	PG-13	114 min	Action & A	This deftly animated take on	2021	8	114		Action & Adventure
330	s329	Movie	Boyka: Un	Todor Cha	Scott Adki	Bulgaria, U	########	2016	R	90 min	Action & A	The fourth film in the action	2021	8	90		Action & Adventure
331	s330	Movie	Catch Me	Steven Spi	Leonardo	United Sta	########	2002	PG-13	142 min	Dramas	An FBI agent makes it his mis	2021	8	142		Dramas
332	s331	Movie	Cloudy wit	Phil Lord, (	Bill Hader,	United Sta	########	2009	PG	90 min	Children &	When inventor Flint Lockwoo	2021	8	90		Children & Family Movies
333	s332	TV Show	Darwin's G	Unknown	Yusuke Ko	Unknown	########	2020	TV-MA	1 Season	Anime Seri	High schooler Kaname activa	2021	8		1	Anime Series
334	s333	Movie	Deep Blue	Renny Har	Thomas Ja	United Sta	########	1999	R	105 min	Action & A	Scientists conduct research c	2021	8	105		Action & Adventure
335	s334	Movie	Ferris Buel	John Hugh	Matthew E	United Sta	########	1986	PG-13	103 min	Classic Mo	After faking an illness to skip	2021	8	103		Classic Movies
336	s335	Movie	Five Feet A	Justin Bald	Haley Lu R	United Sta	########	2019	PG-13	116 min	Dramas, R	A teen with cystic fibrosis sha	2021	8	116		Dramas
337	s336	Movie	Freedomla	Joe Roth	Samuel L.	United Sta	########	2006	R	113 min	Dramas, Th	When a woman reports that	2021	8	113		Dramas
338	s337	TV Show	Friday Nigl	Unknown	Kyle Chan	Unknown	########	2010	TV-14	5 Seasons	TV Dramas	Eric Taylor faces enormous p	2021	8		5	TV Dramas
339	s338	Movie	Good Luck	Mark Helfr	Dane Cook	United Sta	########	2007	R	99 min	Comedies,	Every time Chuck breaks up v	2021	8	99		Comedies
340	s339	TV Show	Hunter X H	Unknown	Megumi H	Japan	########	2014	TV-14	6 Seasons	Anime Seri	To fulfill his dreams of becor	2021	8		6	Anime Series
341	s340	Movie	I missed yo	Mag Hsu, I	Wu Kang-r	Unknown	########	2021	TV-14	96 min	Dramas, In	A workaholic reconnects wit	2021	8	96		Dramas

Figure 6: Cleaned dataset rows 311–340. Full data integrity confirmed across the dataset.

Section 2: Proposed Analytical Technique

2.1 Selected Technique: Naive Bayes Classification

The proposed technique for solving Problem 1 is the **Naive Bayes classifier**, specifically Categorical Naive Bayes given the predominantly categorical feature set. Naive Bayes is a supervised probabilistic classifier based on Bayes' Theorem, computing the posterior probability of a class label given input features under the assumption of conditional independence (Mitchell, 1997).

2.2 Justification

Feature compatibility. The selected features — rating, primary_genre, release_year, year_added, and month_added — are predominantly categorical or low-cardinality discrete numeric variables. Categorical Naive Bayes handles these natively by estimating class-conditional probabilities per category, requiring no normalisation or distance metric selection.

Independence assumption. Correlation analysis of the numeric features shows low inter-feature correlation, and categorical features such as rating and primary_genre are logically independent, a title's maturity rating is determined by content, not genre. This supports the conditional independence assumption.

Class imbalance handling. As identified in Section 1.3, the dataset has a 2.3:1 class imbalance (Movies vs TV Shows). Naive Bayes incorporates class priors directly into its probability framework, making it well-suited to imbalanced data. In Coursework 2, this will be further addressed through stratified train/test splitting (80:20) and class weight adjustment.

Interpretability. Naive Bayes produces probability estimates for each class, enabling transparent communication of predictions to Netflix's strategy team, a key requirement in a business consultancy context.

Comparison with alternatives. KNN performs poorly with mixed categorical and numeric features and scales badly at inference time. Linear Regression is inappropriate for a categorical binary target. K-Means is unsupervised and cannot use the labelled type column. Association Rule Mining is designed for transactional basket data, not classification. Naive Bayes offers the best combination of categorical compatibility, interpretability, and class imbalance awareness for this dataset.

2.3 Selected Attributes

Attribute	Type	Justification
type	Categorical (Binary)	Target variable — Movie (0) or TV Show (1). 2.3:1 imbalance requires stratified splitting in CW2
rating	Categorical	Dominant predictor: TV-MA and TV-14 are strongly associated with TV Shows; R and PG-13 with Movies
primary_genre	Categorical	Genre distributions differ markedly by content type — key discriminating feature
release_year	Numeric (Discrete)	Older content skews Movie; TV Show additions grew post-2016
year_added	Numeric (Discrete)	Netflix acquisition strategy has shifted towards TV Shows over time
month_added	Numeric (Discrete)	Seasonal patterns in content additions may vary by type

Full implementation, including label encoding, stratified splitting, Laplace smoothing, model fitting, and evaluation using accuracy, precision, recall, F1-score, and a confusion matrix, will be carried out in Coursework 2.

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